

Logical Agents and Propositional Logic

CHAPTER 7 AND 8

Moving on...

❑ Atomic search space

- Brittle, had to be modified for every domain
- Enumerating all possible states – impossible for large problems
- Child generator functions, goal test functions are black box to the search algorithm

Knowledge Based Agents

□ Two main components

- Knowledge Base

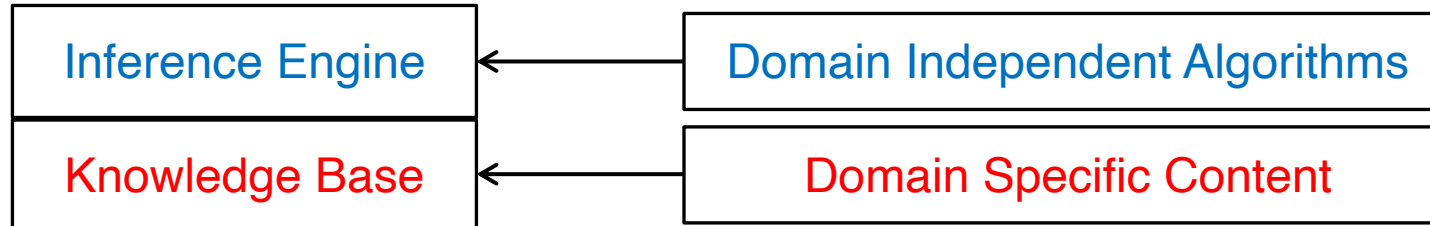
- Set of sentences in a formal language
- Represents some assertion about the world

- Inference Engine

- Mechanism to derive new sentences from old

□ Declarative Approach

- TELL the agent what it needs to know
- ASK itself what to do?



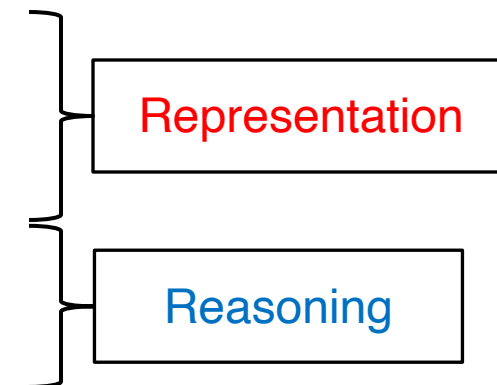
Knowledge Based Agent

```
function KB-AGENT(percept) returns an action
  static: KB, a knowledge base
          t, a counter, initially 0, indicating time

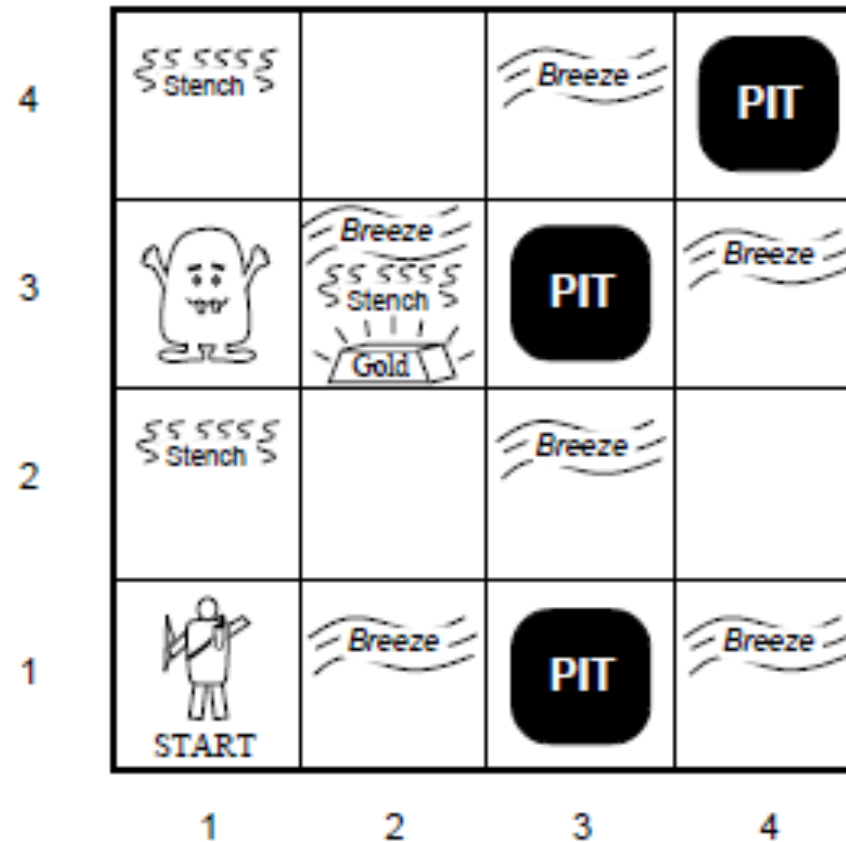
  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
  action ← ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))
  t ← t + 1
  return action
```

□ The agent must

- Represent states and actions
- Incorporate new percepts
- Update internal representations of the world
- Deduce hidden properties of the world
- Deduce appropriate actions



Wumpus World - Example



WW PEAS Description and Characterization

❑ Performance measure

- gold +1000 death -1000
- -1 per step, -10 for arrow

❑ Environment

- Squares adjacent to Wumpus are smelly
- Squares adjacent to pit are breezy
- Glitter iff gold is in the same square
- Shooting kills Wumpus if the agent is facing it
- Shooting uses up the only arrow
- Grabbing picks up the gold if the agent is in the same square
- Climbing in [1,1] gets the agent out of the cave

❑ Sensors

- Stench, Breeze, Glitter, Bump, Scream

❑ Actuators

- TurnLeft, TurnRight, Forward, Grab, Shoot, Climb

❑ Observability: No – only local perception

❑ Deterministic: Yes, outcome is exactly specified

❑ Episodic: No – sequential at the level of actions

❑ Static: Yes, Wumpus, pits and gold do not move

❑ Discrete: Yes

❑ Single Agent: Yes, Wumpus is a natural feature of the environment

Exploring the WW

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

A = Agent
B = Breeze
G = Glitter/Gold
OK = Safe Square
P = Pit
S = Stench
V = Visited
W = Wumpus

Knowledge Base:

- [1,2] and [2,1] are safe

Exploring the WW

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V → OK	2,1 A OK B	3,1 P?	4,1

A = Agent
B = Breeze
G = Glitter/Gold
OK = Safe Square
P = Pit
S = Stench
V = Visited
W = Wumpus

Knowledge Base:

- [1,2] and [2,1] are safe
- [2,1] is breezy
 - Pit in [3,1] or [2,2]?

Exploring the WW

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V A OK	2,1 ← V OK B	3,1 P?	4,1

A = Agent
B = Breeze
G = Glitter/Gold
OK = Safe Square
P = Pit
S = Stench
V = Visited
W = Wumpus

Knowledge Base:

- [1,2] and [2,1] are safe
- [2,1] is breezy
 - Pit in [3,1] or [2,2]?

Exploring the WW

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
1,1	2,1	3,1	4,1

A = Agent
B = Breeze
G = Glitter/Gold
OK = Safe Square
P = Pit
S = Stench
V = Visited
W = Wumpus

- [1,2] is smelly
 - Wumpus in [1,3] or [2,2] or [1,1]
- [1,2] is not breezy
 - Therefore no pit in [2,2]
- [2,1] is not smelly
 - Therefore no Wumpus in [2,2]
- [2,2] has not pit or Wumpus
 - Therefore [2,2] is safe
- [1,2] is smelly and [2,2] and [1,1] are safe
 - Therefore Wumpus is in [1,3]

Knowledge Base:

- [1,2] and [2,1] are safe
- [2,1] is breezy
 - Pit in [3,1] or [2,2]?

Exploring the WW

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
1,1	2,1	3,1	4,1

Knowledge Base:

- [1,2] and [2,1] are safe
- [2,1] is breezy
 - Pit in [3,1] or [2,2]?

A = Agent
 B = Breeze
 G = Glitter/Gold
 OK = Safe Square
 P = Pit
 S = Stench
 V = Visited
 W = Wumpus

- [1,2] is smelly
 - Wumpus in [1,3] or [2,2] or [1,1]
- [1,2] is not breezy
 - Therefore no pit in [2,2]
- [2,1] is not smelly
 - Therefore no Wumpus in [2,2]
- [2,2] has not pit or Wumpus
 - Therefore [2,2] is safe
- [1,2] is smelly and [2,2] and [1,1] are safe
 - Therefore Wumpus is in [1,3]

Exploring the WW

1,4	2,4 P?	3,4	4,4
1,3 W	2,3 OK A B S G	3,3 P?	4,3
1,2 V OK S	2,2 V OK	3,2 OK	4,2
1,1 V OK	2,1 V OK B	3,1 P?	4,1

A = Agent
 B = Breeze
 G = Glitter/Gold
 OK = Safe Square
 P = Pit
 S = Stench
 V = Visited
 W = Wumpus

- [1,2] is smelly
 - Wumpus in [1,3] or [2,2] or [1,1]
- [1,2] is not breezy
 - Therefore no pit in [2,2]
- [2,1] is not smelly
 - Therefore no Wumpus in [2,2]
- [2,2] has not pit or Wumpus
 - Therefore [2,2] is safe
- [1,2] is smelly and [2,2] and [1,1] are safe
 - Therefore Wumpus is in [1,3]

Knowledge Base:

- [1,2] and [2,1] are safe
- [2,1] is breezy
 - Pit in [3,1] or [2,2]?

Exploring the WW

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
1,1	2,1	3,1	4,1

A = Agent
B = Breeze
G = Glitter/Gold
OK = Safe Square
W = Wumpus
P? = Pit

How do we represent rules of the world and the percepts encountered?

Knowledge Base:

- [1,2] and [2,1] are safe
- [2,1] is breezy
 - Pit in [3,1] or [2,2]?

- Wumpus in [1,3] or [2,2] or [1,1]
- [1,2] is not breezy
 - Therefore no pit in [2,2]
- [2,1] is not smelly
 - Therefore no Wumpus in [2,2]
- [2,2] has not pit or Wumpus
 - Therefore [2,2] is safe
- [1,2] is smelly and [2,2] and [1,1] are safe
 - Therefore Wumpus is in [1,3]

Logical Representation

- ❑ Logic - formal languages for representing information such that conclusions can be drawn
- ❑ Sentence – individual piece of knowledge
- ❑ Syntax – What expressions are legal?
 - In the language of arithmetic $x + 2 \geq y$ is a sentence, but $x^2 + y \geq$ is not a sentence
- ❑ Semantics – What the legal expressions mean?
 - $x + 2 \geq y$ is true iff the number $x + 2$ is no less than the number y
 - $x + 2 \geq y$ is true in a world where $x = 5, y = 4$
 - $x + 2 \geq y$ is false in a world where $x = 3, y = 7$

Models

- ❑ Formally structured worlds with respect to which truth can be evaluated
- ❑ We say m is a model of a sentence α if α is true in m
- ❑ $M(\alpha)$ is the set of all models of α
- ❑ Example: $x + 2 \geq y$ is true in a world
 - where $x = 5, y = 4$
 - where $x = 4, y = 4$
 - where $x = 4, y = 6$
 - where $x = 5, y = 3$

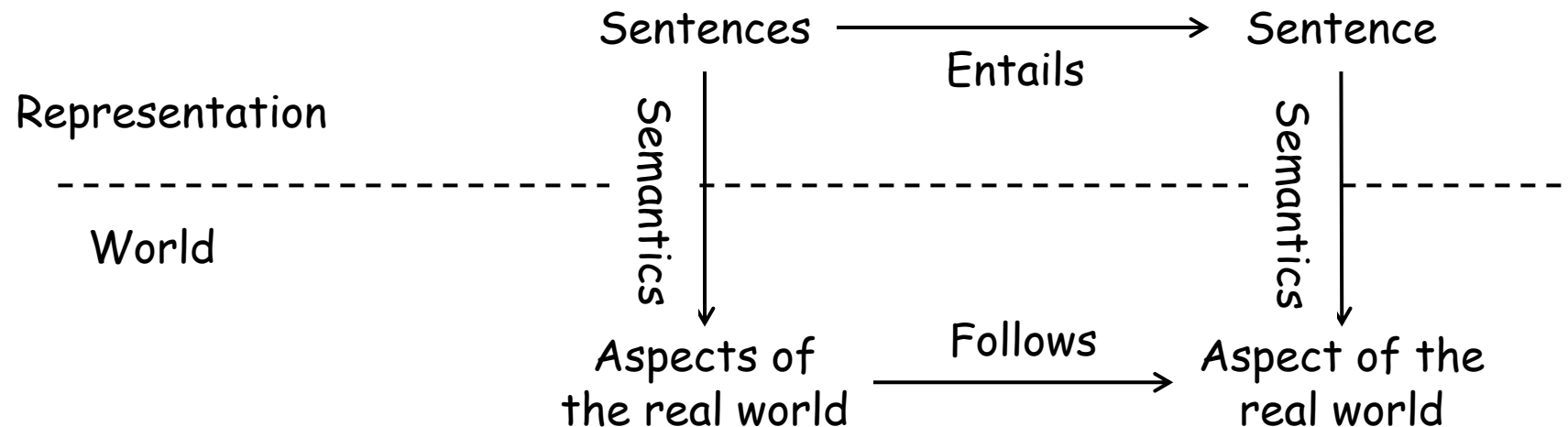
Entailment

□ Means that one thing follows from another:

$$KB \models \alpha$$

□ Knowledge Base KB entails a sentence α iff α is true in all worlds where KB is true

□ Relationship between sentences that is based on semantics



Entailment

- KB:
 - Sun rises in the East
 - Arunachal Pradesh is the Eastern most state of India
- Entails
 - Sun rises first in Arunachal Pradesh in India

- KB:
 - $x + y = 4$
 - $x = 2$
- Entails
 - $4 = x + y$
 - $y = 2$

- KB:
 - Learning is maximum in fun classes
 - AI class is a fun class
- Entails
 - Learning is maximum in AI class

□ KB entails a sentence α iff α is true in all the worlds where KB is also true

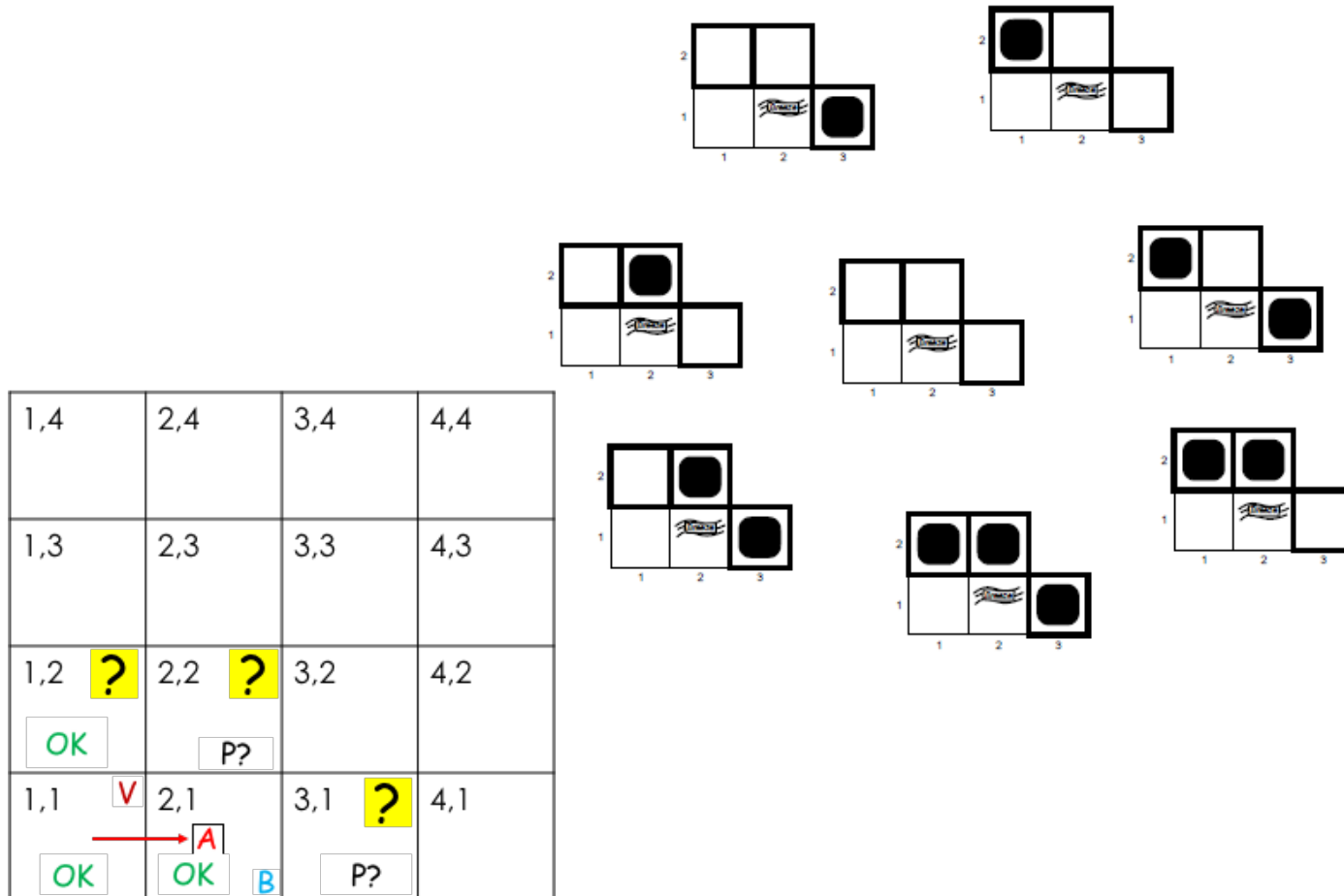
□ $KB \models \alpha$ iff $M(KB) \subseteq M(\alpha)$

Entailment in the WW

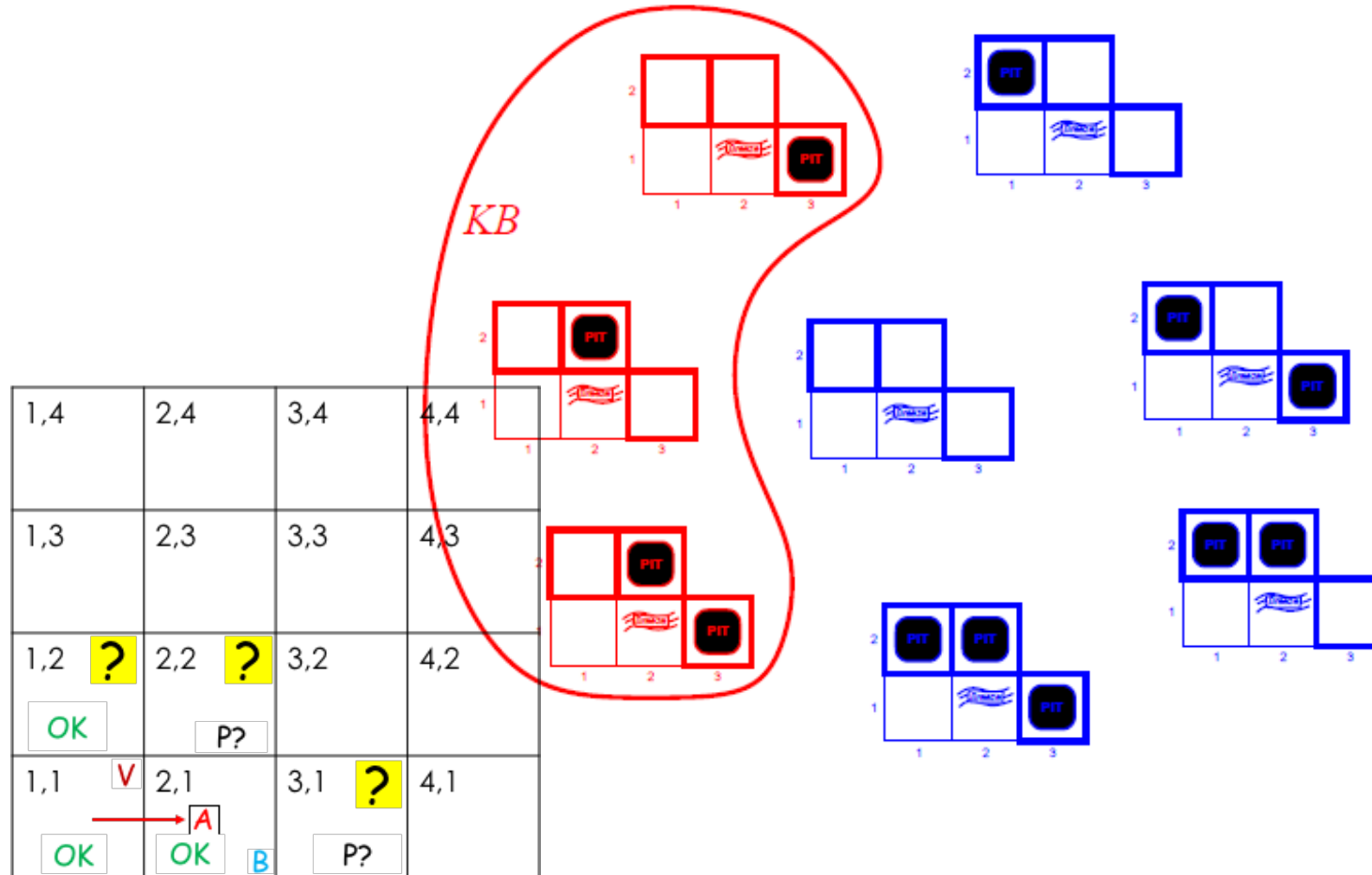
1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 ?	2,2 ? p?	3,2	4,2
1,1 V OK	2,1 A OK B	3,1 ? p?	4,1

- Situation after detecting nothing in [1, 1], moving right, breeze in [2, 1]
- Consider possible models for ?s assuming only pits
- 3 Boolean models = 8 possible models

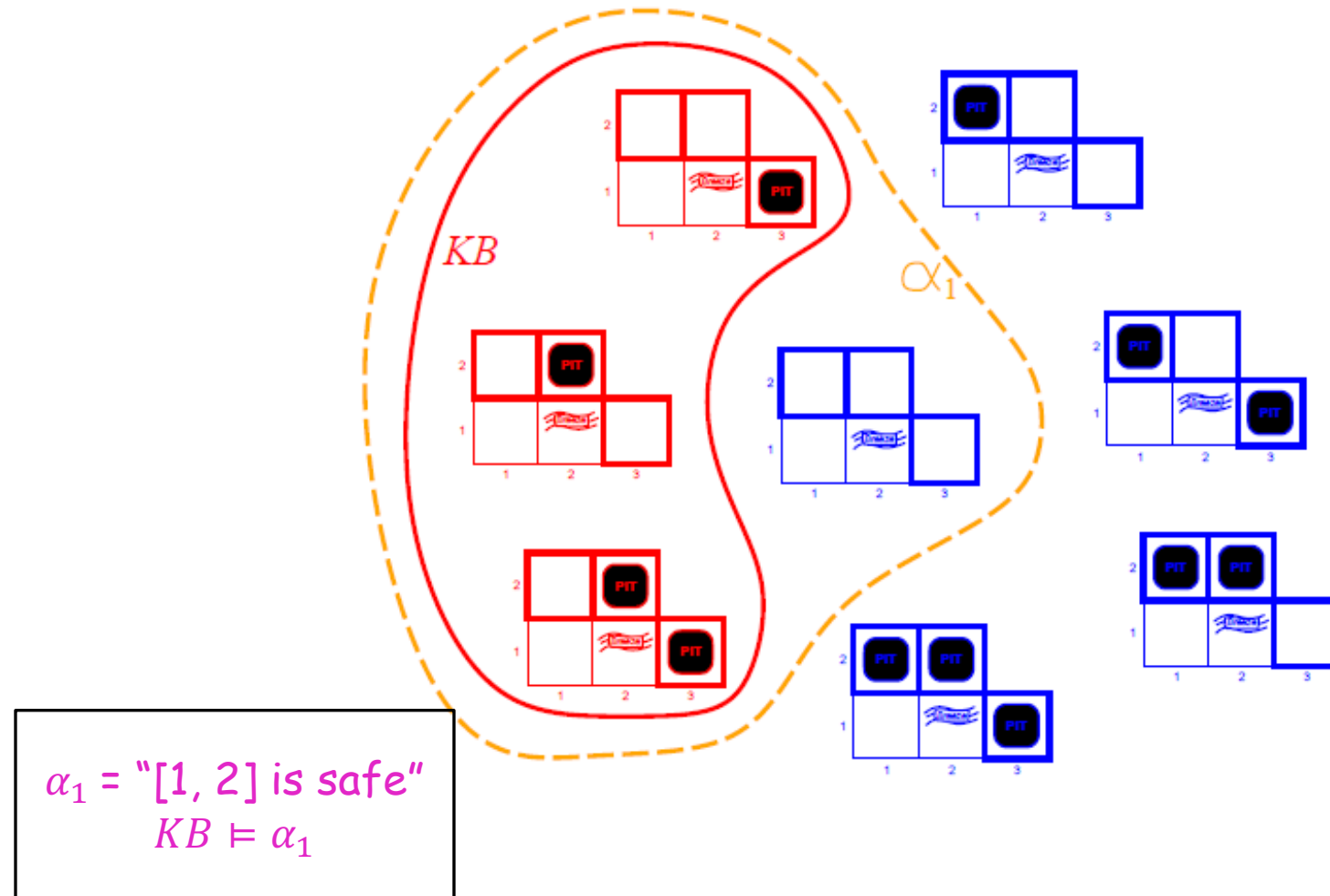
8 possible models



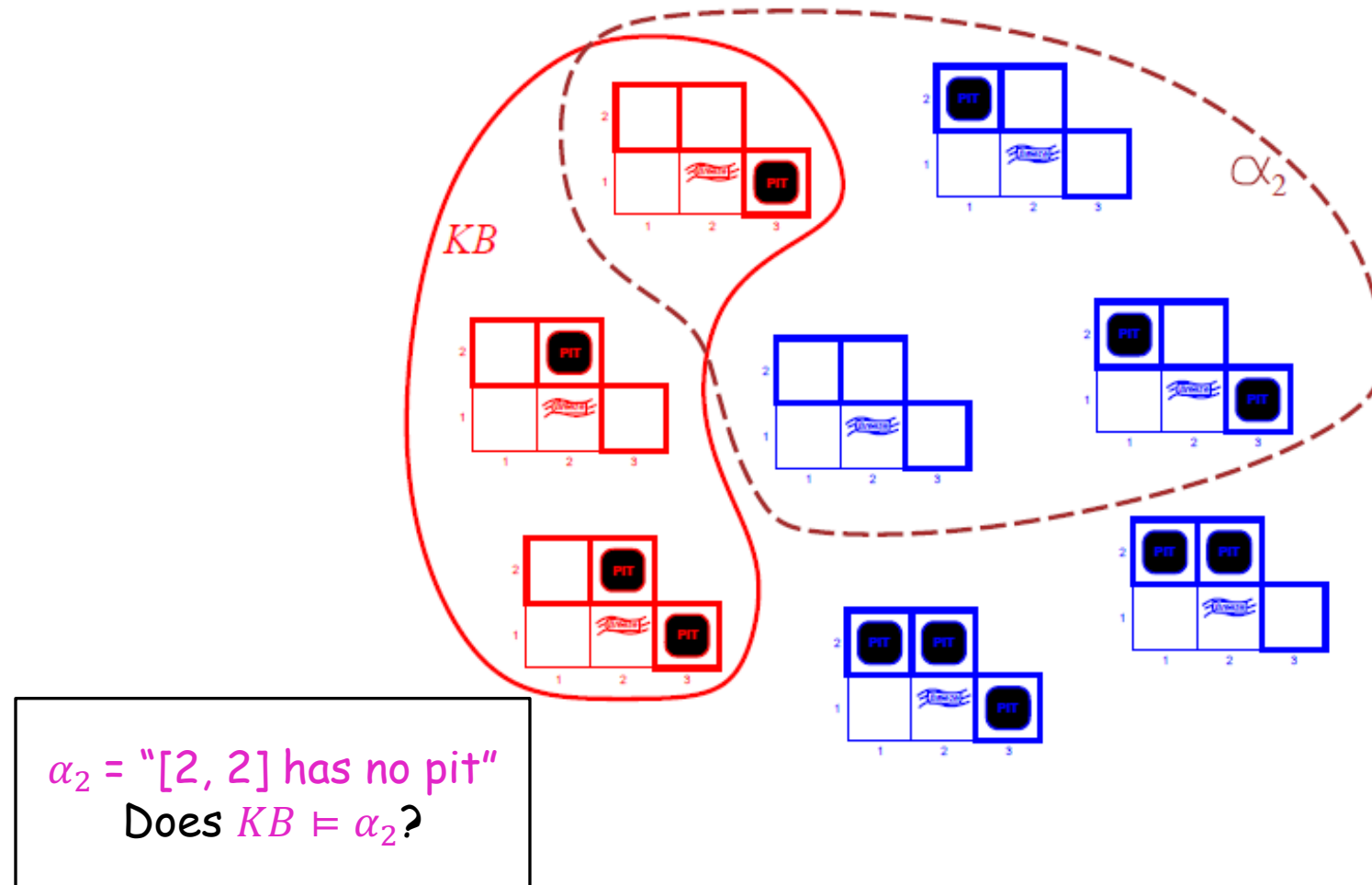
WW Models - KB



WW models – Entailment by model checking



WW models - Entailment



Inference

Feature	Sound	Complete
Meaning	Produces only correct results	Finds all correct results
Error type	No wrong answers	No missing answers
Logical relation	$KB \vdash \alpha \rightarrow \models \alpha$	$\models \alpha \rightarrow \vdash \alpha$

- ❑ $KB \vdash_i \alpha$ – sentence α can be derived from KB by procedure i .
- ❑ All consequences of KB – haystack
- ❑ α is the needle
- ❑ Entailment = needle in the haystack
- ❑ Inference = Finding the needle
- ❑ **Sound**: If the inference algorithm only derives entailed sentences
- ❑ **Complete**: If the algorithm can derive any entailed sentence



Concept	Meaning
Sound	Every needle you pick is real (no fake needles)
Complete	You can find **all** the needles

Your notes use a simple analogy:

Knowledge Base (KB) → the haystack

All logical consequences of KB → things hidden inside the haystack

α (the query) → the needle

So:

Entailment: the needle exists in the haystack

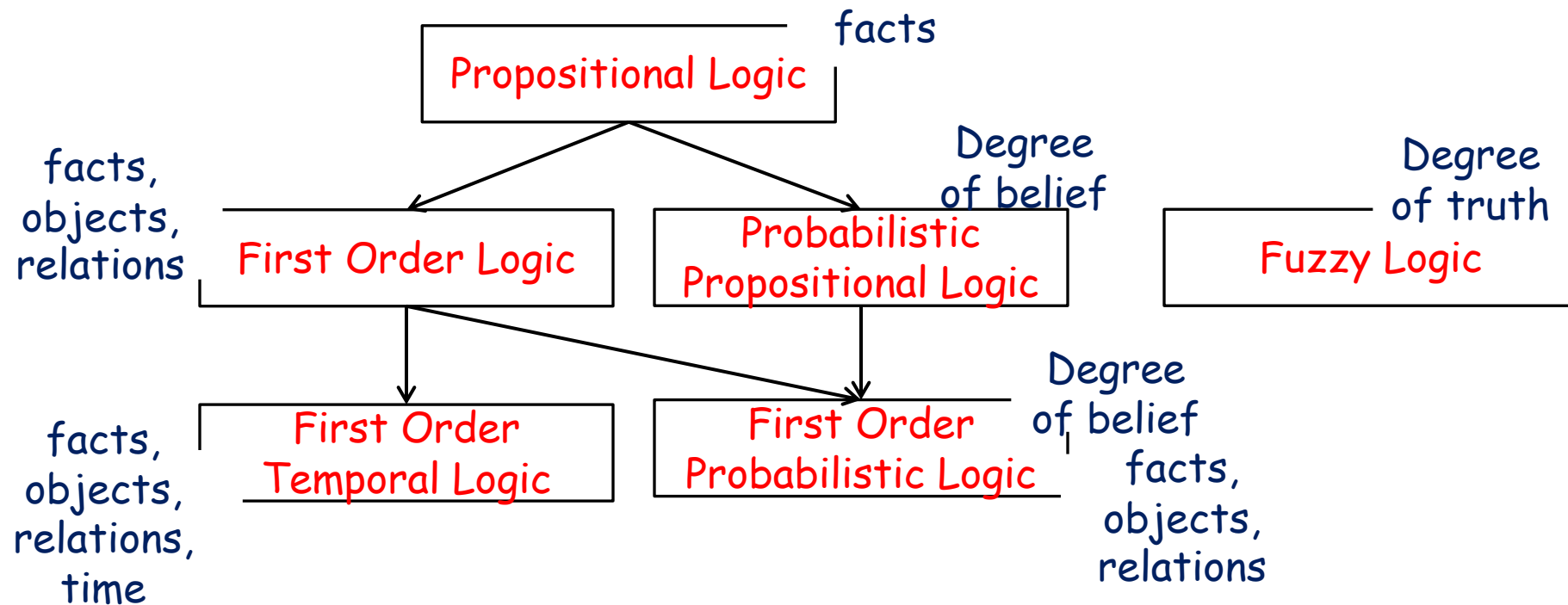
Inference: the process of finding the needle

Types of Logic

❑ Ontological commitments: What exists? –facts, objects, relations, time?

ontological commitment decides what elements the knowledge base can represent.

❑ Epistemological commitments: What states of knowledge?



Propositional Logic - Syntax

- ❑ Simplest Logic – illustrates the basic ideas
- ❑ Atomic sentences – single proposition symbols
 - Can either be *True* or *False* (meaning is fixed)
 - E.g. *A*, *B* that can be either true or false
 - *P*₁₂ or *P*₂₂ etc. used to denote properties of the world
 - *P*₁₂ - pit in square [1,2] – can be true or false in the WW
- ❑ Complex sentences are constructed from simpler sentences recursively using logical connectives
 - Negation - $\neg S$
 - literal
 - Conjunction - $S_1 \wedge S_2$
 - Disjunction - $S_1 \vee S_2$
 - Implication - $S_1 \Rightarrow S_2$
 - Premise (antecedent)
 - Conclusion (consequent)
 - Biconditional - $S_1 \Leftrightarrow S_2$

Propositional Logic – Syntax Backus-Naur Form

$Sentence \rightarrow AtomicSentence \mid ComplexSentence$

$AtomicSentence \rightarrow True \mid False \mid P \mid Q \mid R \mid \dots$

$ComplexSentence \rightarrow (Sentence) \mid [Sentence]$

$\mid \neg Sentence$

$\mid Sentence \wedge Sentence$

$\mid Sentence \vee Sentence$

$\mid Sentence \Rightarrow Sentence$

$\mid Sentence \Leftrightarrow Sentence$

OPERATOR PRECEDENCE : $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Propositional Logic - Semantics

- A model specifies *true/false* for each proposition symbol.
- Rules for evaluating truth w.r.t. a model *m*:

$\neg S$	is true iff	S	is false		
$S_1 \wedge S_2$	is true iff	S_1	is true and	S_2	is true
$S_1 \vee S_2$	is true iff	S_1	is true or	S_2	is true
$S_1 \Rightarrow S_2$	is true iff	S_1	is false or	S_2	is true
	i.e., is false iff	S_1	is true and	S_2	is false
$S_1 \Leftrightarrow S_2$	is true iff	$S_1 \Rightarrow S_2$	is true and	$S_2 \Rightarrow S_1$	is true

P_{12}	P_{22}	P_{31}	$\neg P_{12} \wedge (P_{22} \vee P_{31})$
<i>false</i>	<i>false</i>	<i>true</i>	$= \neg \text{false} \wedge (\text{false} \vee \text{true})$
			$= \text{true} \wedge \text{true} = \text{true}$

Truth Tables for connectives

S_1	S_2	$\neg S_1$	$S_1 \wedge S_2$	$S_1 \vee S_2$	$S_1 \Rightarrow S_2$	$S_1 \Leftrightarrow S_2$
false	false	true	false	false	true	true
false	true	true	false	true	true	false
true	false	false	false	true	false	false
true	true	false	true	true	true	true



Psst... here's a useful tip:
 $S_1 \Rightarrow S_2$ is equivalent to $\neg S_1 \vee S_2$
Prove it!

Exercise: Write an algorithm for computing the truth value of a propositional logic sentence s given a model m

Wumpuspositional Logic

□ Syntax for WW sentences

- P_{ij} is true if there is a pit in $[i, j]$
- B_{ij} is true if there is breeze in $[i, j]$

□ Knowledge Base:

- True Statements

$$\neg P_{11}, \neg B_{11}, B_{21}$$

- Properties of the Wumpus World

- “Pits cause breeze in adjacent squares”

$$B_{11} \Leftrightarrow (P_{21} \vee P_{12})$$

$$B_{21} \Leftrightarrow (P_{11} \vee P_{22} \vee P_{31})$$

- A square is breezy iff there is an adjacent pit



1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 OK	2,1 V → A OK B	3,1 P?	4,1

Inference Proof/Techniques

□ Model Checking

A sequence of inference rule applications that derive a goal from the knowledge base.

- Truth Table enumeration (exponential in n)
- Efficient Backtracking algorithms
 - e.g. David-Putnam-Logemann-Loveland (DPLL)
- Local Search Algorithms (sound but incomplete)
 - e.g. randomized hill climbing (WalkSAT)

□ Application of inference rules

- Generate new sentences from old in a sound way
- Proof – a sequence of inference applications
- Use inference rules as actions/operators in a standard search algorithm
- Typically require translation of sentence into a normal form.

Inference by Truth Table Enumeration

	B_{11}	B_{21}	P_{11}	P_{12}	P_{21}	P_{22}	P_{31}	KB	$\neg P_{12}$
Model 1	false	false	false	false	false	false	false	false	true
Model 2	false	false	false	false	false	false	true	false	true
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	false	true	false	false	false	false	false	false	true
	false	true	false	false	false	false	true	true	true
	false	true	false	false	false	true	false	true	true
	false	true	false	false	false	true	true	true	true
	false	true	false	false	true	false	false	false	true
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	true	true	true	true	true	true	true	false	false

In all models in which KB is true, $\neg P_{12}$ is also true.

Therefore $KB \models \neg P_{12}$

$$\neg P_{11}, \neg B_{11}, B_{21}$$

$$B_{11} \Leftrightarrow (P_{21} \vee P_{12})$$

$$B_{21} \Leftrightarrow (P_{11} \vee P_{22} \vee P_{31})$$

Inference by Truth Table Enumeration

B_{11}	B_{21}	P_{11}	P_{12}	P_{21}	P_{22}	P_{31}	KB	P_{22}
false	false	false	false	false	false	false	false	false
false	false	false	false	false	false	true	false	false
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
false	true	false	false	false	false	false	false	false
false	true	false	false	false	false	true	true	false
false	true	false	false	false	true	false	true	true
false	true	false	false	false	true	true	true	true
false	true	false	false	true	false	false	false	false
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
true	true	true	true	true	true	true	false	true

P_{22} is false in a model in which KB is true

Therefore, $KB \not\models P_{22}$

$$\neg P_{11}, \neg B_{11}, B_{21}$$

$$B_{11} \Leftrightarrow (P_{21} \vee P_{12})$$

$$B_{21} \Leftrightarrow (P_{11} \vee P_{22} \vee P_{31})$$

Inference by Enumeration

```
function TT-ENTAILS?( $KB, \alpha$ ) returns true or false  
  inputs:  $KB$ , the knowledge base, a sentence in propositional logic  
          $\alpha$ , the query, a sentence in propositional logic  
   $symbols \leftarrow$  a list of the proposition symbols in  $KB$  and  $\alpha$   
  return TT-CHECK-ALL( $KB, \alpha, symbols, []$ )
```

```
function TT-CHECK-ALL( $KB, \alpha, symbols, model$ ) returns true or false  
  if EMPTY?( $symbols$ ) then  
    if PL-TRUE?( $KB, model$ ) then return PL-TRUE?( $\alpha, model$ )  
    else return true  
  else do  
     $P \leftarrow$  FIRST( $symbols$ );  $rest \leftarrow$  REST( $symbols$ )  
    return TT-CHECK-ALL( $KB, \alpha, rest, \text{EXTEND}(P, \text{true}, model)$ ) and  
          TT-CHECK-ALL( $KB, \alpha, rest, \text{EXTEND}(P, \text{false}, model)$ )
```

- Depth first exploration
- Complexity - $O(2^n)$ n – symbols

- Sound - Yes
- Complete - Yes